Heat Sealing of LDPE, HS-BOPP, PET/LDPE, PET/MPET/LDPE and BOPP/MBOPP Laminates in Bag-Making Machines: Parameter Windows, Machine Evolution and Future Benchmarks

K Vijay Anand
Chennai, India
haivijayanand@gmail.com

***Abstract*—Heat-seal quality on bag-making machines is set by the interplay of jaw temperature, dwell time, pressure and the thickness of the sealant layer. This paper compiles starting sealing parameters for five common flexible-packaging structures —LDPE mono film, heat-sealable BOPP, PET/LDPE, PET/MPET/LDPE and BOPP/MBOPP—across their usual thickness ranges, and explains the trends through a heat-conduction model. A PET outer layer roughly doubles the usable sealing window (35–45 °C against 20–30 °C) because it does not melt below about 255 °C. Required dwell grows with the square of the conduction path, so doubling sealant thickness quadruples dwell at a given jaw temperature. The paper also traces bag-making machines from cam-driven mechanical lines to multi-axis servo machines rated at 240–500 cycles/min, and reviews how recyclability regulation and mono-material PE and PP laminates will narrow sealing windows again. A short test protocol for fixing the real window on a production line is given.**

***Index Terms*—Heat sealing, flexible packaging, LDPE, BOPP, PET laminate, metallised film, bag-making machine, seal window, dwell time, mono-material packaging.**

I. Introduction

Flexible bags and pouches are closed by heat sealing: two sealant faces are pressed between heated jaws until the interface melts, the polymer chains inter-diffuse, and the seal sets on cooling. In every structure considered here only the inner layer seals; the outer layers decide how hot the jaw may run before the film shrinks, burns through or delaminates (Table I).

PET (12 µm) gives stiffness, a print surface and heat resistance. Metallised PET adds an oxygen and moisture barrier but cracks if overheated or flexed. BOPP is clear, stiff and inexpensive, but because it is oriented it shrinks when heated near its melting range of about 160–165 °C. The sealant—LDPE, or a thin PP copolymer skin on BOPP—is the only layer that melts and fuses.

The aims of this paper are to (i) give practical starting settings for each structure and thickness, (ii) explain the trends physically, (iii) summarise defect causes, and (iv) place sealing in the context of past, present and future bag-making machines.

TABLE I
Structures, Sealing Layers and Jaw-Temperature Limits

Structure	Build (µm)	Sealing layer	Initiation (°C)	Upper limit (°C)	Upper limit set by
LDPE	25–200	LDPE	105–115	170–190	Burn-through, edge thinning
HS-BOPP	15–40	PP co-/terpolymer skin, 0.5–1.5 µm	105–125	145–155	Oriented core shrinks
PET/LDPE	12/30–150	LDPE	130–150	200–210	PET shrink, adhesive failure
PET/MPET/LDPE	12/12/40–120	LDPE	140–160	200–210	Adhesive failure, metal cracking
BOPP/MBOPP	15–20/18–25	MBOPP heat-seal skin	115–130	150–160	OPP shrink, seal distortion

II. Sealing Fundamentals

A. Temperature, Time and Pressure

Jaw temperature and dwell together set the interface temperature and therefore whether a seal forms at all. Pressure only has to bring the faces into full contact and expel air; above about 0.2–0.5 MPa it adds little, and excess pressure squeezes molten sealant from the seal edge and thins it.

B. Thickness and Heat Conduction

The time for heat to reach the interface scales with the square of the conduction path L:

$$t \approx L^2/\alpha, \quad \alpha_{\mathrm{LDPE}} = k/(\rho\, c_p) \approx 0.16 \text{ mm}^2/\text{s} \tag{1}$$

where $k \approx 0.33$ W/m·K, $\rho \approx 920$ kg/m³ and $c_p \approx 2300$ J/kg·K. L is one film thickness when both jaws are heated and two when only one is, as on most impulse sealers. Real dwells are 10–50 times the pure-conduction value because of the PTFE cover, contact resistance and the latent heat of melting, but the square-law trend holds.

C. Effect of a PET Outer Layer

PET acts as both heat shield and heat barrier: it lets the jaw run 30–50 °C hotter without damage, but its 12 µm adds conduction lag, so PET laminates need a higher jaw temperature than plain LDPE of the same sealant thickness.

D. Hot Tack and Seal Window

Hot tack is the strength of a still-molten seal just after the jaws open; it matters when product drops onto the bottom seal. Cold seal strength, measured after cooling (ASTM F88), is what the customer sees. The seal window runs from the lowest

temperature giving a film-tearing seal to the highest before shrinkage, thinning or delamination. A wide window tolerates temperature variation along the bar and changes in line speed.

III. Sealing Parameters and Trends

Table II lists starting settings for a constant-heat bar sealer. For a single-side impulse sealer, dwell should be increased by 30–50 % and a cooling time about equal to the heat time allowed. All values are typical industry starting points for trials, not measurements on a particular film or machine.

TABLE II
Starting Sealing Parameters for a Constant-Heat Bar Sealer, Both Jaws Heated, 5–10 mm Seal

Structure	Build (µm)	Jaw temp. (°C)	Dwell (s)	Pressure (MPa)	Cold seal strength (N/15 mm)
LDPE	30	125–140	0.4–0.7	0.15–0.25	5–7
LDPE	50	135–150	0.6–1.0	0.20–0.30	8–12
LDPE	75	145–160	0.9–1.5	0.20–0.30	12–16
LDPE	100	150–170	1.2–2.0	0.25–0.35	15–20
LDPE	150	165–185	2.0–3.0	0.30–0.40	20–28
HS-BOPP	15	110–125	0.3–0.5	0.30–0.50	1.5–2.5
HS-BOPP	20	115–130	0.4–0.7	0.30–0.50	2–3
HS-BOPP	30	120–135	0.5–0.9	0.30–0.50	2.5–4
HS-BOPP	40	125–140	0.7–1.1	0.35–0.55	3–5
PET/LDPE	12/40	150–165	0.3–0.6	0.30–0.40	12–18
PET/LDPE	12/60	160–175	0.5–0.8	0.30–0.45	18–26
PET/LDPE	12/90	170–185	0.8–1.2	0.35–0.50	25–35
PET/LDPE	12/120	180–195	1.0–1.5	0.40–0.50	30–42
PET/MPET/LDPE	12/12/50	165–180	0.5–0.8	0.35–0.45	15–22
PET/MPET/LDPE	12/12/75	175–190	0.8–1.2	0.35–0.50	22–32
PET/MPET/LDPE	12/12/100	185–200	1.0–1.6	0.40–0.55	28–38
BOPP/MBOPP	15/18	120–135	0.3–0.6	0.35–0.50	2–3.5
BOPP/MBOPP	20/20	125–140	0.4–0.7	0.35–0.55	2.5–4.5
BOPP/MBOPP	20/25	130–145	0.5–0.8	0.40–0.60	3–5

Fig. 1 shows that every structure needs a hotter jaw as it gets thicker, by about 0.3–0.4 °C per µm of LDPE. The PET laminates sit highest yet stay clear of their own damage limit; the BOPP structures sit lowest and are bounded by core shrinkage only 10–20 °C above their range. For BOPP the sealing layer is the thin skin, so thicker film raises settings only modestly.

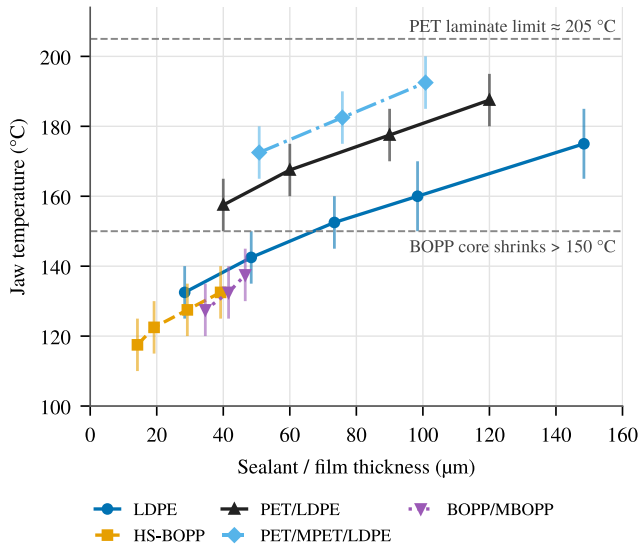


Fig. 1. Starting jaw temperature versus thickness (LDPE layer for PET laminates, total film for BOPP structures). Dot: range mid-point; bar: recommended range; dashed lines: upper limits.

Fig. 2 applies (1), calibrated to 0.8 s at 142 °C for 50 µm LDPE. Raising temperature shortens dwell with diminishing return, and for LDPE burn-through caps temperature near 180 °C. Thick films therefore cannot be sealed within one fast machine cycle at any safe temperature, which is why high-speed pouch machines repeat the seal over two to four stations.

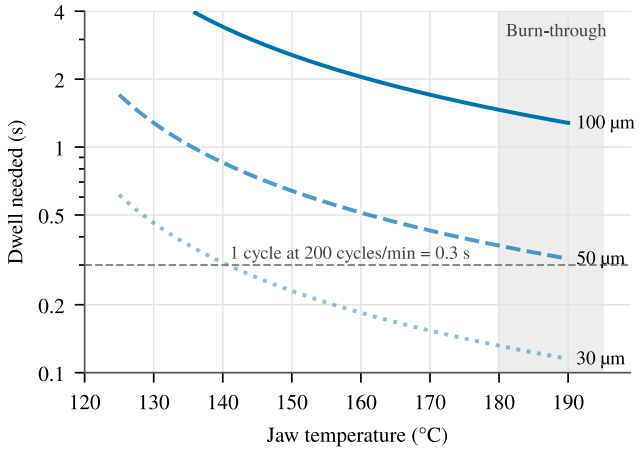


Fig. 2. Dwell needed for a film-tearing LDPE seal versus jaw temperature, from (1) with $t = k'L^2/(T - 110)$; both jaws heated; log scale.

Fig. 3 shows the three zones of every seal curve: weak sticking below initiation, a plateau where the seal fails by film tear, and a decline as the film thins, shrinks or delaminates. Peak strengths differ widely—about 10 N/15 mm for 50 µm LDPE, 3–4 for BOPP structures and 23–28 for PET laminates

—so the curves are normalised to compare window widths directly.

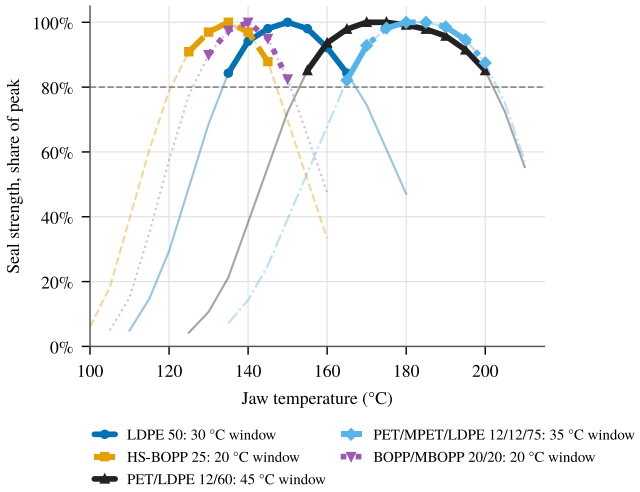


Fig. 3. Illustrative cold seal strength, normalised to each structure's peak, versus jaw temperature at 0.6–0.8 s dwell. Bold segment: usable window (≥ 80 % of peak).

IV. Pressure, Jaw Design and Sealing Method

The regulator reading is cylinder air pressure, not seal pressure. Two 50 mm bore cylinders at 4 bar give about 1570 N; over a 600 × 10 mm seal this is about 0.26 MPa. The figure must be recalculated whenever seal length or width changes. Table III compares sealing methods.

TABLE III
Sealing Methods on Bag-Making Machines

Method	Heat input	Best suited to	Typical settings
Constant-heat bar	Both jaws held hot for a set dwell	PET laminates, BOPP structures; 3-side-seal and stand-up pouches	Table II; dwell split over 2–4 stations plus a cooling station
Impulse band	Thin band pulsed hot, cooled under pressure	Thick LDPE, bottom-seal bags, gussets	Heat 0.5–3 s, cool about equal
Hot knife (side weld)	Heated blade cuts and fuses a bead	Thin LDPE, HDPE, PP bags; HS-BOPP garment bags	Knife ≈ 300–450 °C, 0.05–0.2 s
Serrated jaws	Constant-heat jaws with fine teeth	BOPP/MBOPP, HS-BOPP centre-seal pouches	130–160 °C, 0.2–0.5 s
Ultrasonic	Cold horn, 20–40 kHz; friction heats interface	Contaminated or heat-sensitive seals	≈ 80–200 ms; tool stays cold

Seal width of 5–10 mm is typical; wider seals improve pinhole tolerance, not peel strength. Temperature uniformity along the bar (±3 °C, verified by contact thermocouple) matters more than the set point. One soft, silicone- or PTFE-backed jaw evens pressure over gusset and zipper steps. PTFE covers are needed for LDPE and HS-BOPP; PET-outer laminates usually run on bare jaws. Cooling bars after the hot station let thick seals set flat.

V. Defects and Troubleshooting

Most seal faults arise from an interface that is too cold (weak, peelable seals) or an outer layer that is too hot (shrinkage, burn-through, delamination). Table IV maps symptoms to causes.

TABLE IV
Seal Defects, Causes and First Corrective Action

Symptom	Most affected	Likely cause	First fix
Seal peels apart cleanly	All	Interface too cold or dwell too short	+5–10 °C or +0.1–0.2 s
Holes or thin line at seal edge	LDPE	Too hot or too much pressure	Lower temperature and pressure; check PTFE
Puckered, shrunk seal	HS-BOPP, BOPP/MBOPP	Oriented core near its melt range	−10 °C, longer dwell; serrated jaws
Wrinkled seal, curled bag	PET laminates	PET shrinking above ≈ 200 °C	Lower temperature, add dwell
Layers separate at seal	PET/LDPE, PET/MPET/LDPE	Adhesive overheated or uncured	Lower temperature; cure PU 48–72 h
Metallised layer cracked	PET/MPET/LDPE, BOPP/MBOPP	Excess heat or crimp pressure	Reduce temperature and crimp depth
Weak seal at high temperature	Laminates	Wrong face, slip bloom, contamination	Sealant faces in; check slip; clean seal
Leaks at gusset or zipper	All pouches	Thickness step lowers local pressure	Soft or profiled jaw, more pressure
Strength varies along bar	All	Uneven bar temperature or parallelism	Map bar with thermocouple, re-level
Seal opens on product drop	LDPE, BOPP	Low hot tack	Cool under pressure; mLLDPE sealant

VI. Evolution of Bag-Making Machines

The decisive change has been decoupling seal time from machine speed (Table V). On a cam-driven machine the dwell is a fixed fraction of the cycle: doubling speed halves dwell, so operators raised jaw temperature to compensate—the main source of burn-through and shrink defects on older lines. Separately driven servo seal heads hold the programmed dwell

at any speed, so Table II can be applied independent of line speed on a modern machine.

TABLE V
EVOLUTION OF BAG-MAKING MACHINES

Era	Drive and control	Sealing technology	Speed (cycles/min)
1940s–50s	Manual, foot-operated	Hot wands and bars; impulse sealing from mid-1940s [1]	Manual
1960s	Main motor, cams, clutch-brake feed	Constant-heat PTFE-covered bars; first automatic PE bag makers	40–80*
1970s	Mechanical, in-line with extruder [2]	Moving-carriage seal bars; first laminate pouch machines	60–120*
1980s–90s	Variable drives, photocell registration, PLC	Mature wicketers and side-weld lines [3]	100–250
2000s	Multi-axis servo, touch HMI	Seal time independent of speed; multi-station sealing	150–300
2010s–20s	Full servo, auto changeover, remote diagnostics	Servo seal heads, closed-loop jaw temperature, multi-lane	240–500

*Approximate typical values, not sourced ratings.

VII. CURRENT BENCHMARKS

Top machines run 240–360 cycles/min for laminate pouches and 400–500 cycles/min for thin PE bags (Table VI). The gap reflects the longer dwell laminate seals need, bought on pouch machines by repeating the seal over several stations. Rated speeds are maxima; actual output depends on material, bag size and seal time.

TABLE VI
CURRENT MACHINE BENCHMARKS (RATED MAXIMA)

Machine class	Structures	Rated speed	Example
PE wicketer	LDPE, LLDPE 20–60 µm	400–450+ cycles/min	[7], [9]
Side-weld bag maker	PP, LDPE, HDPE 10–100 µm	350–450 strokes/min	[11]
3-side-seal pouch	PET and BOPP laminates	240; up to 360 claimed	[4], [5]
Centre-seal pouch	BOPP/MBOPP, PET laminates	240–250 cycles/min	[13]
Multi-lane pouch	Laminates	≈ 1,200 pouches/min (4 lanes)	[6]
Flat-bottom pouch	Laminates, gussets, zippers	Up to 150 pouches/min	[12]

A modern line should also provide seal time set independent of speed [8], closed-loop jaw temperature with per-zone alarms, print registration that stops production on a missing eye mark, and quick changeover—flat-bottom pouch lines quote under 20 min [10].

VIII. FUTURE DIRECTIONS

Recyclability regulation, not speed, will drive the next change. The EU Packaging and Packaging Waste Regulation applies from 12 August 2026 and requires all packaging on the EU market to be recyclable from 2030, graded A, B or C at 95 %, 80 % and 70 % recyclable by weight, with grade C phased out by 2038 [14], [15]. Packaging must also be recycled at scale, which is expected to push out multi-layer mixed-polymer laminates [16].

In all-PE laminates the print web is machine-direction-oriented PE, which cannot tolerate PET-level jaw temperatures; low-temperature plastomer sealants are used so the seal forms before the MDO layer distorts [17]. Mono-material windows are much narrower, so time and temperature must be matched precisely [18]. Table VII summarises emerging technologies.

TABLE VII
EMERGING TECHNOLOGIES AND THEIR EFFECT ON SEALING

Technology	Effect on sealing	Status
All-PE / all-PP laminates	Window narrows to BOPP-like range; low-SIT sealant needed [17], [18]	In production
Ultrasonic sealing	Cold tool, 80–200 ms, seals through contamination [19]; lower strength on some laminates [20]	FFS; emerging on bag makers
Low-mass heated jaws	Hold set temperature at high cycle rates	Emerging
In-line seal inspection	Every seal checked, not sample peel tests	Emerging
Recipe libraries	Seal window stored and recalled per job	On servo machines

IX. RECOMMENDED SEAL-WINDOW TEST

1. Condition film 24 h at 23 °C, 50 % RH; confirm PU adhesive cure (48–72 h).
2. Fix pressure and dwell at Table II mid-values; verify seal pressure from cylinder force and seal area.
3. Map jaw surface temperature at five points; correct deviations beyond ±3 °C.
4. Step jaw temperature in 5 °C increments from 20 °C below to 20 °C above the table range; five seals per step.
5. Peel-test 15 mm strips at 300 mm/min after 24 h (ASTM F88); record peak force and failure mode.
6. Define the window as temperatures giving film-tear failure (or ≥ 80 % of plateau strength) with no visual defect.
7. Repeat at dwell ±0.2 s; set production at the window centre.
8. Leak-test finished bags, focusing on gussets and zipper ends.

X. CONCLUSION

PET-outer laminates are the easiest structures to seal and the most forgiving; plain LDPE is easy but bounded at both ends; BOPP structures are least forgiving because the oriented core shrinks close to the sealing temperature. Thicker sealant needs hotter jaws and disproportionately longer dwell, so dwell should be set for the thickest sealant a job uses. Interface pressure of 0.2–0.5 MPa should be verified by calculation, not read from the gauge. Servo machines remove the historical speed-versus-dwell compromise and should be run at the window centre. Recyclable mono-material structures will narrow the window again; building in-house seal-window test capability now is the best preparation.

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